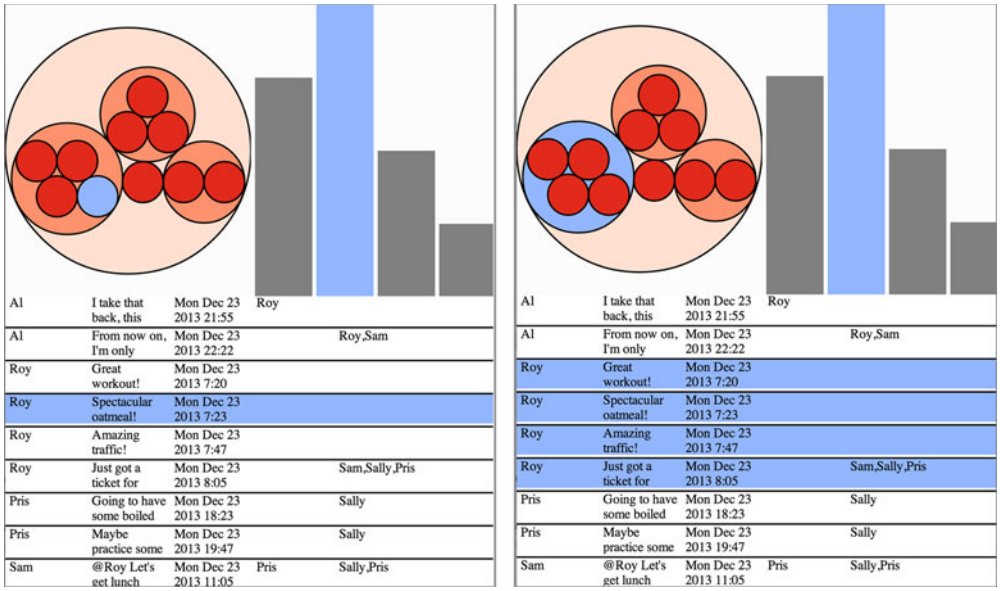


Figure 9.7 Cross-highlighting in action. The same code provides both results shown if we mouseover an individual data row or level 2 circle (right), highlighting the corresponding bar on the bar chart that represents that nested data. It also highlights the data rows that correspond to the nested elements in the data for the level 1 circles or bars when we mouseover those (left).

Obviously, the same principles could be applied to click events, or you could reintroduce the modal pop-up we used in chapter 3 to expose more functionality. But rather than add more interactivity, I'll switch gears and introduce a new control: the brush.

9.3 Brushing

The brush component, `d3.svg.brush()`, is like the axis component because it creates SVG elements when called (typically by a `<g>` element). But it's also like the zoom behavior because brush has interactions that update a data element that you can use for interactivity. Brushes are valuable interactive components that allow users to intuitively slice up their data. For our dashboard, we'll add a brush that lets users show tweets only from particular times of the day. You not only need to learn how to use the brush component and understand the brush events that come with the component, but also, because we'll base our activity on when tweets are made, you need to understand D3's built-in scale for dealing with time: `d3.time.scale`.

9.3.1 Creating the brush

A brush in D3 takes a scale and creates a region where the user can select part of that scale by clicking and dragging. First, we need a new div into which we'll place our brush, and so we need to update our page to include a `#brush` div in between our chart divs and our spreadsheet div.